MASTER FUNCTIONAL SKILLS (MFS) · WORKED SOLUTIONS ·
MasterFunctionalSkills.co.uk · AQA Maths Level 2 Specialist
Full Marks (100% Pass) · Verified by Tutor Abdul · November 2021 - Paper 2

AQA

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

M F S 0 1 2 0 2 6

Surname MASTER

Forename(s) FUNCTIONAL SKILLS

Candidate signature Master Functional Skills

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Candidate: Master Functional Skills
Date: 20/10/2026 · Tutor Abdul

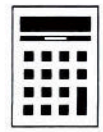
# Functional Skills Level 2 MATHEMATICS

## Paper 2 Calculator

Time allowed: 1 hour 30 minutes

**Materials**

For this paper you must have:
- a calculator
- mathematical instruments.



**Instructions**
- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- State the units of your answer where appropriate.

**Information**
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 60.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.
- If your calculator does not have a π button, take the value of π to be 3.142

**Advice**

In all calculations, show clearly how you work out your answer.

| For Examiner's Use | |
|---|---|
| Question | Mark |
| 1–6 | |
| 7 | |
| 8 | |
| 9 | |
| 10 | |
| **TOTAL** | |

N O V 2 1 8 3

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Do not write outside the box

## Section A

Answer **all** questions in the spaces provided.

**1** The probability of an event happening is $\frac{3}{5}$

Circle the probability of the event **not** happening.

**[1 mark]**

1 $\frac{2}{5}$ $\frac{3}{5}$ 0

**2** $x = 5$ and $y = -3.2$

Work out the value of $x + y^2$

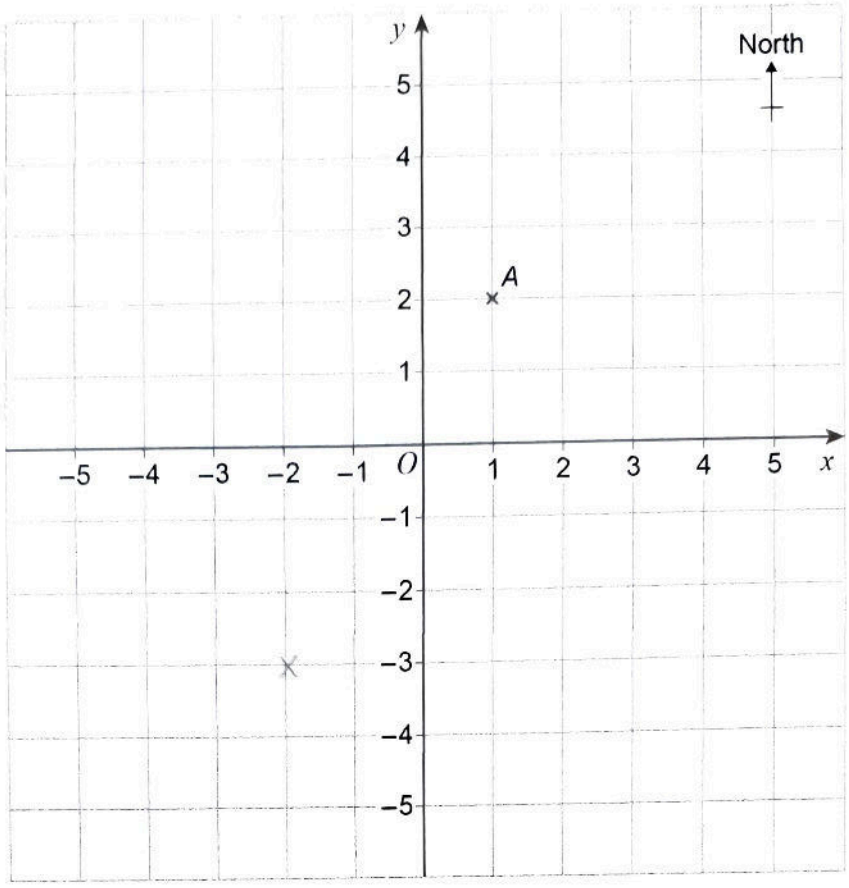
**[2 marks]**

$5 + (-3.2)^2 = 15.24$

Answer 15.24

Do not write outside the box

**3**




*A* is the point (1, 2)

*B* is the point

3 squares West of *A*

*and*

5 squares South of *A*.

Work out the coordinates of *B*.

**[2 marks]**

Answer ( −2 , −3 )

**Turn over ▶**

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**4** In the boxes, write the number that is

3 more than −8 | −5 |

4 less than −2 | −6 |

halfway between −7 and 5 | −1 |

**[3 marks]**

**5** Write the ratio 144 : 90 in its simplest form.

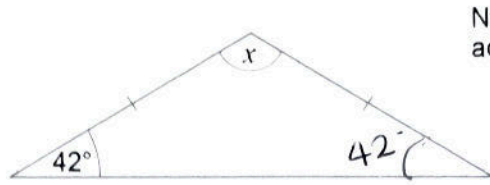
**[2 marks]**

144 : 90
16 : 10
8 : 5

Answer 8 : 5

Do not write outside the box

**6** Here is an isosceles triangle.




Work out the size of angle $x$.

**[2 marks]**

$180 - 2(42) = 96$

Answer 96 °

12

**Turn over for Section B**

Do not write outside the box

## Section B

Answer **all** questions in the spaces provided.

**7 Energy**

Jim's house has gas central heating.

**7 (a)** The charges for the gas Jim uses are shown in the table.

| | |
|---|---|
| Cost per unit of gas (kWh) | 2.92p |
| Standing charge per day | 39.6p |

In November, Jim uses 1976 kWh of gas.

Use **approximations** to estimate Jim's November gas payment.

Give your answer in pounds.

**[5 marks]**

1976 x 2.92 =

approximately → 2000 x 3 = 6000

per day approx 40 x 30 days

= 1200

Total = 6000 + 1200

= 7200p

= £72

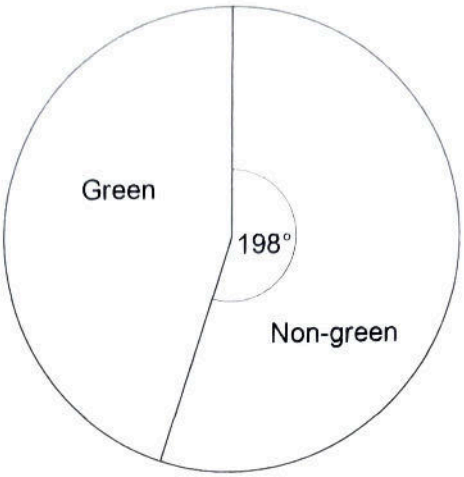
Answer £ 72

Do not write outside the box

**7 (b)** Jim's current gas company produces 43% of their gas from green sources.

He sees this pie chart for another gas company, MixGas.

**Gas source**




Jim wants to use the company with a greater proportion of gas from green sources.

Should Jim change to MixGas?

You **must** show your working.

**[3 marks]**

$360° - 198° = 162°$

proportion of total circle $\frac{162}{360} = 0.45 \times 100 = 45\%$

$45\% > 43\%$

Yes Jim should change to MixGas

**Question 7 continues on the next page**

**Turn over ▶**

Do not write outside the box

**7 (c)** Jim is thinking of buying solar panels.

The price of these panels has decreased by 27% since 2016

Jim would need to pay £4803.40 for the panels now.

How much would the panels have cost in 2016?

**[3 marks]**

100 - 27 = 73

4803.40 ÷ 73 x 100 = 6580

Answer £ 6580

11

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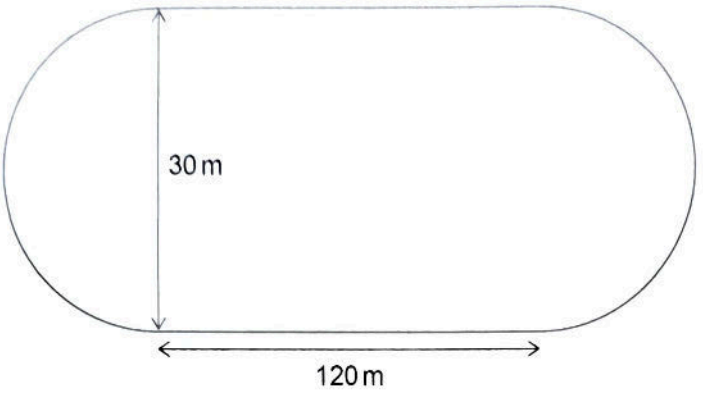
**8 Dog**

Ben has a dog called Pluto.

**8 (a)** Ben takes Pluto for a walk.

They walk **4 laps** of a sports track.

The sports track is the shape of two semi-circles joined by two straight sections.




Not drawn accurately

Ben has a target to walk 5000 m each day.

What percentage of his daily target has Ben completed on this walk?

**[5 marks]**

area of a hemisphere = $\frac{1}{2}\pi d$

$= \frac{1}{2}\pi (30) = 47.12$

2 hemispheres so $47.12 \times 2 = 94.25$

Total = $94.25 + 120 + 120 = 334.2$

4 laps = $334.2 \times 4 = 1337$

% = $1337 \div 5000 \times 100$

Answer 27 %

**Question 8 continues on the next page**

Do not write outside the box

**8 (b)** Pluto needs to eat 420 grams of food each day, in the ratio

wet food : dry food = 2 : 5

Ben buys a 12 kilogram bag of **dry** food.

How many days of **dry** food does the bag provide?

**[4 marks]**

420 grams 2 : 5

420 ÷ 7 = 60g

wet = 60 x 2 = 120g

dry = 60 x 5 = 300g

12kg = 12000g

12000 ÷ 300 = 40 days

Answer 40 days

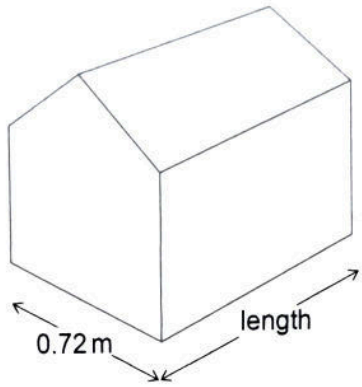
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**8 (c)** In his garden, Ben has a rectangular fenced area for Pluto.

The fenced area measures 2.4 m by 2.25 m

Ben builds a kennel for Pluto which covers $\frac{1}{6}$ of the fenced area.

Here is a sketch of the kennel.




The base of the kennel is a rectangle.

Work out the length of the kennel.

**[4 marks]**

fenced area:
$2.4\text{m} \times 2.25\text{m} = 5.4\text{m}^2$

kennel:
$\frac{1}{6}$ of $5.4\text{m}^2 = 0.9\text{m}^2$

$0.9\text{m}^2 = 0.72\text{m} \times \text{length}$

$\text{length} = \frac{0.9}{0.72}$

$= 1.25\text{m}$

Answer 1.25 m

13

**Turn over for the next question**

Do not write outside the box

**9 Fundraising**

A charity organises an annual sponsored swim.

**9 (a)** Ahmed swam 50 lengths of a pool that is 25 metres long.

The swim took him 38 minutes.

Ahmed says,

"My average speed was more than 0.7 metres per **second**."

Is Ahmed correct?

You **must** show your working.

**[4 marks]**

50 lengths x 25m = 1250 m

38 mins = 2280 seconds

speed in metres per second = $\frac{1250}{2280}$

$= 0.548 ms^{-1}$

No his speed was less than $0.7 ms^{-1}$

Do not write outside the box

**9 (b)** Ahmed collected £35 for his swim and gave it to the charity.

$\frac{6}{7}$ of this money was donated by taxpayers.

Using Gift Aid, the charity claimed an extra 25p for every £1 donated by a taxpayer.

How much money did the charity receive, **including** Gift Aid, from Ahmed's swim?

**[3 marks]**

$35 \times \frac{6}{7} = 30$

£30 by taxpayers

25p extra per £1

£30 x 0.025 = 7.5

total = £35 + £7.5 = £42.50

Answer £ 42.50

**Question 9 continues on the next page**

**Turn over ▶**

Do not write outside the box

**9 (c)** This year, 20 people completed the sponsored swim.

The table shows information about the sponsor money collected.

| Money (£$x$) | Frequency | Mid-point | frequency x midpoint |
|---|---|---|---|
| $0 < x \leqslant 10$ | 7 | 5 | 35 |
| $10 < x \leqslant 20$ | 4 | 15 | 60 |
| $20 < x \leqslant 30$ | 6 | 25 | 150 |
| $30 < x \leqslant 40$ | 1 | 35 | 35 |
| $40 < x \leqslant 50$ | 2 | 45 | 90 |
| | Total = 20 | | 370 |

Last year, the mean amount collected per swimmer was £14.85

The charity organiser says,

"This year's mean amount is **more than** £3 higher than last year's."

Show your working to support this statement.

**[5 marks]**

7 x 5 = 35

4 x 15 = 60

6 x 25 = 150

1 x 35 = 35

2 x 45 = 90

370

370 ÷ 20 = 18.5 mean

difference between this year and last year:

£18.5 - £14.85 = £3.65

∴ Yes, more than £3 higher

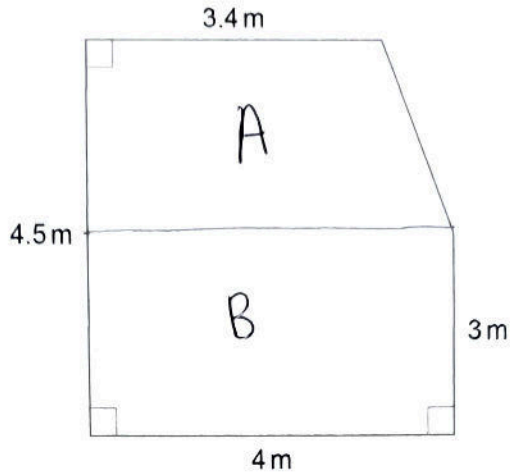
12

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**10 Conservatory**

Molly is building a conservatory.

**10 (a)** Here is a plan view of the conservatory.




Not drawn accurately

The conservatory floor will be made of concrete.

The concrete floor will be 0.1 m deep.

What volume of concrete will Molly need for the floor?

State the units of your answer.

**[6 marks]**

Area = area A + area B

A : $\frac{1}{2}(3.4 + 4) \times (4.5 - 3)$

$\frac{1}{2}(7.4) \times 1.5 = 5.55$

B : $3 \times 4 = 12$

Total area = $12 + 5.55 = 17.55$

volume = $17.55 \times 0.1 = 1.755 m^3$

Answer $1.755 m^3$

**Question 10 continues on the next page**

**Turn over ▶**

Do not write outside the box

**10 (b)** Molly has a scale drawing of the side elevation of the conservatory.

The scale of the drawing is 3 : 50

The height of the conservatory on the scale drawing is 23.4 centimetres.

Is the actual height of the conservatory **less than** 4 metres?

You **must** show your working.

**[3 marks]**

Scale height = 23.4cm

3 : 50

23.4 ÷ 3 = 7.8cm

7.8 x 50 = 390cm

390cm = 3.9m

Yes the actual height is less than 4m (3.9m).

Do not write outside the box

**10 (c)** Molly wants to build a brick wall as part of the conservatory.

It would take Molly 12 hours to build the wall by herself.

Molly pays 2 builders to work with her to build the wall.

Molly pays each builder £14.73 per hour.

How much will Molly pay each builder for their work?

Assume that Molly and the builders work at the same rate.

**[3 marks]**

12 hours ÷ 3 = 4 hours with 3 builders (molly + 2)

4 x £14.73

= £58.92

Answer £ 58.92

12

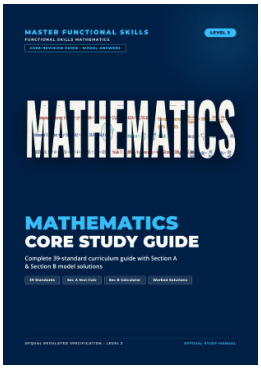
**END OF QUESTIONS**

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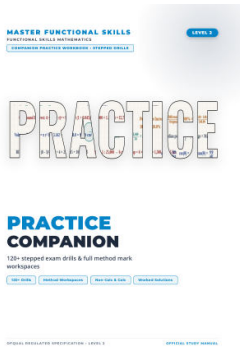



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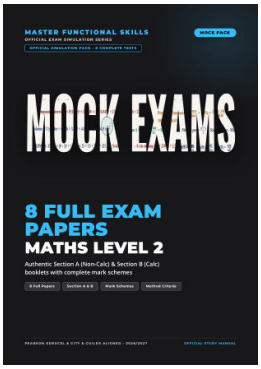



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